

DIFFERENTIAL EQUATIONS



SYNOPSIS

- **Differential Equation:** An equation involving one dependent variable and its derivatives w.r.t to one or more independent variables is called a differential equation.

Eg: 1. $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = x^2$

2. $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = u$

- **Types of differential equation:**

- i) **Ordinary differential equation:** A differential equation involving derivatives with respect to single independent variable is called ordinary differential equation.

Eg: $\frac{dy}{dx} = x^3 + \sin x$

- ii) **Partial differential equation:** A differential equation involving at least two independent variables and partial derivatives with respect to either of these independent variables is called a partial differential equation.

Eg: $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u$

Order and Degree of a differential equation :

Definition: The order of an ordinary differential equation is defined to be the order of the highest order derivative occurring in the equation.

Definition: The degree of an ordinary differential equation is defined to be the exponent(index) of the highest order derivative occurring in the equation provided the equation is made free from radical signs and fractional powers as far as the derivatives concerned (i.e., when the equation is expressed as a polynomial equation of the derivatives). Every differential equation has a certain order but need not have a degree.

W.E-1: Find the order and degree (if defined) of the following differential equations:

Sol: i) $y = 1 + \left(\frac{dy}{dx}\right) + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \dots$

The given differential equation can be rewritten

as $y = e^{dy/dx} \Rightarrow \frac{dy}{dx} = \log y$

Hence, its order is 1 and degree 1

ii) $\left(\frac{d^3y}{dx^3}\right)^{2/3} = \frac{dy}{dx} + 2 \Rightarrow \left(\frac{d^3y}{dx^3}\right)^2 = \left(\frac{dy}{dx} + 2\right)^3$

Hence, its order is 3 and degree 2

iii) $\left(\frac{d^2y}{dx^2}\right) = x \log\left(\frac{dy}{dx}\right)$

It's order is obviously 2.

Since, the given differential equation cannot be written as a polynomial in all the differential coefficients, the degree of the equation is not defined

- **Solution of the differential equation:** A relation between the variables without derivatives which satisfy the given differential equation is called a solution of the given differential equation.

e.g.1: $y = a \sin x$, $y = b \cos x$ are solutions of

differential equation $\frac{d^2y}{dx^2} + y = 0$

e.g.2: $y = ae^x$ and $y = be^{2x}$ are solutions of

differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

- **Types of Solutions:**

- i) **Formation of differential equation:** Let the given equation be

$f(x, y, c_1, c_2, \dots, c_n) = 0 \dots (1)$

Differentiating n times with respect to x , we get n more equations. By eliminating the arbitrary constants $c_1, c_2, \dots, c_n$ we get the differential equation of order n .

ii) General solution: The solution which contains as many arbitrary constants as the order of the differential equation is called general solution of the differential equation.

e.g.1: $y = a \sin x + b \cos x$ (a, b are arbitrary

constants) is general solution of $\frac{d^2 y}{dx^2} + y = 0$

e.g.2: $y = ae^x + be^{2x}$ (a, b are arbitrary constants)

is general solution of $\frac{d^2 y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

iii) Particular solution: Solution obtained by giving particular values to the arbitrary constants in the general solution of the given equation.

e.g.1: $y = 5e^x + 6xe^x$ is particular solution of

differential equation $\frac{d^2 y}{dx^2} - \frac{2dy}{dx} + y = 0$

W.E-2: The differential equation whose general solution is given by

$$y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5} \quad \text{where}$$

c_1, c_2, c_3, c_4, c_5 are arbitrary constants, is

Sol: We can write

$$y = A \cos(x + B) - Ce^x$$

where

$$A = c_1 + c_2, B = c_3 \text{ and } C = c_4 e^{c_5}$$

$$\frac{dy}{dx} = -A \sin(x + B) - Ce^x$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -A \cos(x + B) - Ce^x$$

$$\Rightarrow \frac{d^2 y}{dx^2} + y = -2Ce^x$$

$$\Rightarrow \frac{d^3 y}{dx^3} + \frac{dy}{dx} = -2Ce^x = \frac{d^2 y}{dx^2} + y$$

$$\Rightarrow \frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$$

W.E-3: The differential equation of all parabolas whose axes are parallel to the axis of y , is

Sol: Such parabolas are given by

$$(x - h)^2 = 4a(y - k),$$

where h, k, a are three arbitrary constants,

on differentiating again w.r.to x , $(x - h) = 2a \frac{dy}{dx}$

on differentiating again, $1 = 2a \frac{d^2 y}{dx^2} \Rightarrow \frac{d^2 y}{dx^2} = \frac{1}{2a}$

on differentiating once again, $\frac{d^3 y}{dx^3} = 0$

This is the required differential equation.

➤ **Types of differential equations:**

- i) Variable separable
- ii) Homogeneous equations
- iii) Non-homogeneous equation
- iv) Linear equations
- v) Bernoulli's equations

i) Variable separable method:

Its general form is

$$\Rightarrow \frac{dy}{dx} = \frac{f(x)}{g(y)} \quad (\text{or}) \quad \frac{dy}{dx} = \frac{g(y)}{f(x)}$$

Where $f(x), g(y)$ are continuous functions.

$$\Rightarrow g(y) dy = f(x) dx \quad (\text{or}) \quad \frac{dx}{f(x)} = \frac{dy}{g(y)}$$

By integration we get the solution.

$$\text{e.g. } y(1 - ay) dx - (x + a) dy = 0$$

Note: If the equation is of the form

$\frac{dy}{dx} = f(ax + by + c)$; put $ax + by + c = z$ then convert into variable separable form.

$$\text{e.g.: } \frac{dy}{dx} + 1 = e^{x+y}$$

$x + y = z$, reduce to the form of variable separable differential equation

W.E-4: Solution of the Differential Equation

$$y - x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right) \text{ is}$$

Sol: We have, $y - x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right)$

$$\Rightarrow y dx - x dy = ay^2 dx + a dy$$

$$\Rightarrow y(1 - ay) dx = (x + a) dy$$

$$\Rightarrow \frac{dx}{x+a} - \frac{dy}{y(1-ay)} = 0 \text{ Integrating, we get}$$

$$\log(x+a) - \log y + \log(1-ay) = \log c$$

$$\Rightarrow \log \frac{(a+x)(1-ay)}{y} = \log c$$

$$\Rightarrow (x+a)(1-ay) = cy$$

ii) Homogeneous Function: A function $f(x, y)$ is said to be homogeneous function in x, y of degree n if $f(kx, ky) = k^n f(x, y)$ where n is a constant.

➤ **Homogeneous differential equation :**

If $f(x, y)$ and $g(x, y)$ are homogeneous functions of same degree then the D.E.

$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$ is called Homogeneous differential equation

Working rule :

1. Substitute $y = vx$ where v is a function of 'x' and

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

2. Reduce to the form of variable separable differential equation.

eg.1: $\frac{dy}{dx} = \frac{ax+by}{cx+dy}$ **e.g.2:** $\frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$ (or)

Working rule :

1. Substitute $x = vy$ where v is a function of 'y' and

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

2. Reduce to the form of variable separable differential equation

W.E-5: Solution of the differential equation

$$x \sin \frac{y}{x} dy = \left(y \sin \frac{y}{x} - x \right) dx \text{ is}$$

Sol: We have $\frac{dy}{dx} = \frac{y \sin \frac{y}{x} - x}{x \sin \frac{y}{x}}$ Put $y = Vx$

$$\text{So that } \frac{dy}{dx} = V + x \frac{dV}{dx}$$

$$\text{Hence, } V + x \frac{dV}{dx} = \frac{V \sin V - 1}{\sin V} = V - \frac{1}{\sin V}$$

$$\Rightarrow x \frac{dV}{dx} = -\frac{1}{\sin V} \Rightarrow \int \frac{dx}{x} + \int \sin V dV = c$$

$$\Rightarrow \log x - \cos V = c \Rightarrow \log x = \cos \frac{y}{x} + c$$

W.E-6: Solve $\left(1 + e^{\frac{x}{y}} \right) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) dy = 0$

Sol: $\left(1 + e^{\frac{x}{y}} \right) dx = \left(\frac{x}{y} - 1 \right) e^{\frac{x}{y}} dy$; $\frac{dx}{dy} = \frac{(x-y)e^{\frac{x}{y}}}{y \left(1 + e^{\frac{x}{y}} \right)}$

$$\text{put } x = vy \text{ ; } \frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$v + y \frac{dv}{dy} = \frac{(v-1)e^v}{1+e^v}; \quad y \frac{dv}{dy} = \frac{ve^v - e^v}{1+e^v} - v$$

$$y \frac{dv}{dy} = \frac{ve^v - e^v - v - ve^v}{1+e^v}; \quad y \frac{dv}{dy} = \frac{-(v+e^v)}{1+e^v}$$

$$\frac{1+e^v}{v+e^v} dv = \frac{-1}{y} dy; \quad \int \frac{1+e^v}{v+e^v} dv = -\int \frac{1}{y} dy$$

$$\left[\because \int \frac{f'(x)}{f(x)} dx = \log(f(x) + c) \right]$$

$$\log(v + e^v) = -\log y + c; \text{ but } v = \frac{x}{y}$$

$$\log \left(\frac{x}{y} + e^{\frac{x}{y}} \right) = -\log y + c$$

iii) Non-homogeneous differential equation:

General form is $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$

where $a_1, b_1, c_1, a_2, b_2, c_2$ are constants

Case i: If $b_1 = -a_2$

Working Rule: Arrange equation in the form

$$a_2xdy + b_2ydy + c_2dy = a_1xdx + b_1ydx + c_1dx$$

$$\Rightarrow a_2(xdy + ydx) + b_2ydy + c_2dy$$

$$-a_1xdx - c_1dx = 0$$

$\Rightarrow$ Solution is

$$a_2xy + b_2\frac{y^2}{2} + c_2y - a_1\frac{x^2}{2} - c_1x = k$$

Case ii: If $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ ($b_1 \neq -a_2$)

Working rule: Put $a_1x + b_1y = t$ and reduce to the form of variable separable DE

Case iii: If $b_1 \neq -a_2$ and $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

Working rule:

1. Put $x = X + h, y = Y + k$ to convert given DE as homogenous D.E.
2. Take $a_1h + b_1k + c_1 = 0, a_2h + b_2k + c_2 = 0$ and find h, k

iv) Linear differential equations:

Type(i): General form $\frac{dy}{dx} + P(x) \cdot y = Q(x)$.

Then $e^{\int P(x)dx}$ is the integrating factor and the

solution is $y \cdot e^{\int P(x)dx} = \int Q(x) \cdot e^{\int P(x)dx} dx + c$

Type (ii): General Form: $\frac{dx}{dy} + P(y)x = Q(y)$.

Then integrating factor is $e^{\int P(y)dy}$. Then solution is

$$x \cdot e^{\int P(y)dy} = \int Q(y) \cdot e^{\int P(y)dy} dy + c$$

Note: Points to consider

- Here the term $e^{\int p dx}$ and $e^{\int p dy}$ which converts the left hand expression of the equation into a perfect differential is called an integrating factor. In short it is written as **I.F**

W.E-7: Solve $\frac{dy}{dx} + 2y = \cos x$.

Sol: It is a linear equation of the form $\frac{dy}{dx} + Py = Q(x)$

where $P = 2$ and $Q = \cos x$

$$\text{Then IF} = e^{\int p dx} = e^{\int 2 dx} = e^{2x}$$

Hence, the general solution is

$$y(IF) = \int Q(IF) dx; y \cdot e^{2x} = \int e^{2x} \cos x dx + C$$

$$y \cdot e^{2x} = \frac{e^{2x}}{5} [2 \cos x + \sin x] + C$$

W.E-8: Solution of the differential equation

$$(x + 2y^3) \frac{dy}{dx} = y \text{ is}$$

Sol: We have $(x + 2y^3) \frac{dy}{dx} = y$

$$\Rightarrow y \frac{dx}{dy} = x + 2y^3 \Rightarrow \frac{dx}{dy} - \frac{1}{y}x = 2y^2$$

which is a linear differential equation in x

$$\text{I.F} = e^{\int P dy} = e^{-\int \frac{1}{y} dy} = e^{-\log y} = e^{\log \left(\frac{1}{y}\right)} = \frac{1}{y}$$

$$\therefore \text{The solution is } x \cdot \frac{1}{y} = \int 2y^2 \cdot \frac{1}{y} dy + c$$

$$\Rightarrow x \cdot \frac{1}{y} = y^2 + c \Rightarrow x = y(c + y^2)$$

v) Bernoulli's differential equation:

Type 1: General form $\frac{dy}{dx} + P(x) \cdot y = Q(x) \cdot y^n$,

$n \in R$ is called Bernoulli's equation in y . Dividing

by y^n and substituting $\frac{1}{y^{n-1}} = Z$ convert into linear differential equation.

Type 2: General form $\frac{dx}{dy} + P(y)x = Q(y)x^n$.

$n \in R$ is called Bernoulli equation in x .

W.E-9: Solve $(y \log x - 1)y dx = x dy$

Sol: The given differential equation can be written as

$$x \frac{dy}{dx} + y = y^2 \log x$$

$$\text{Let } \frac{1}{y} = v \Rightarrow -\frac{1}{y^2} \cdot \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{So that } \frac{dv}{dx} - \frac{1}{x} v = \frac{1}{x} \log x$$

which is the standard linear differential equations, with

$$P = -\frac{1}{x}, Q = -\frac{1}{x} \log x$$

$$IF = e^{\int -1/x dx} = e^{-\ln x} = e^{\ln x^{-1}} = \frac{1}{x}$$

The solution is given by

$$v \cdot \frac{1}{x} = \int \frac{1}{x} \left(-\frac{1}{x} \log x \right) dx = \int \frac{\log x}{x^2} dx$$

$$= \frac{\log x}{x} - \int \frac{1}{x} \cdot \frac{1}{x} dx = \frac{\log x}{x} + \frac{1}{x} + C$$

$$\Rightarrow v = 1 + \log x + Cx = \log(ex) + Cx$$

$$\text{or } \frac{1}{y} = \log(ex) + Cx \text{ or } y \{ \log(ex) + Cx \} = 1$$

Orthogonal trajectory :

- Any curve which cuts every member of curve at right angle is called an orthogonal trajectory of the family.

Procedure for finding the orthogonal trajectory:

- Let $f(x, y, c) = 0$ be the equation, where 'c' is an arbitrary parameter.
- Differentiate the given equation w.r.t. x and then eliminate 'c'.
- Substitute $-\frac{dx}{dy}$ for $\frac{dy}{dx}$ in the equation obtained in (ii)
- By solving the D.E. we get the required orthogonal trajectory, for example, each straight line passing through the origin i.e. $y = mx$ is an orthogonal trajectory of the family of circles $x^2 + y^2 = a^2$

W.E-10: Find the orthogonal trajectory of the circles $x^2 + y^2 - ay = 0$

Sol: Here $x^2 + y^2 - ay = 0$ (i)

Differentiating, we get $2x + 2yy_1 - ay_1 = 0$

$$\Rightarrow a = \frac{2(x + yy_1)}{y_1} \text{ substituting 'a' in Eq(i) we get}$$

$$x^2 + y^2 - \frac{2(x + yy_1)}{y_1} y = 0$$

$$\Rightarrow (x^2 - y^2)y_1 - 2xy = 0$$

This is the differential equation of the family of circles given in Eq.(i)

∴ The differential representing the orthogonal trajectory is obtained by replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$

$$\text{i.e., } -(x^2 - y^2) \frac{dx}{dy} - 2xy = 0$$

$$\Rightarrow 2xy dy - y^2 dx = -x^2 dx$$

$$\Rightarrow \frac{xd(y^2) - y^2 dx}{x^2} = -dx \Rightarrow d\left(\frac{y^2}{x}\right) = -dx$$

Integrating both the sides, we get

$y^2 + x^2 = Cx$, is required family of orthogonal trajectories.

➤ Geometrical Applications:

We also use differential equations for finding family of curves for which some conditions involving the derivatives are given.

Equation of tangent at a point (x,y) on the curve $y = f(x)$ is given by

$$Y - y = \frac{dy}{dx} (X - x).$$

At the x-axis, $Y=0$, so that $X = x - y/\frac{dy}{dx}$ and the y-axis $X=0$, so that

$$Y = y - x \frac{dy}{dx}. \text{ Hence the length}$$

of intercept of tangent on the x-axis is $x - y/\frac{dy}{dx}$ and

the y-axis is $y - x \frac{dy}{dx}$ similar information can be obtained for normals by writing its equation as

$$(y - y) \frac{dy}{dx} + (X - x) = 0$$

W.E-11: A normal is drawn at a point $P(x,y)$ of a curve. It meet the x -axis at Q . If PQ is of constant length k , then show that the differential equation describing such curves is $y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$. Find the equation of such a curve passing through $(0,k)$.

Sol: Equation of the normal at a point $P(x,y)$ is given by

$$Y - y = -\frac{1}{dy/dx}(X - x) \dots \dots \dots (1)$$

Let the point Q at the x -axis be $(x_1, 0)$. from (1). we

$$\text{get } y \frac{dy}{dx} = x_1 - x \dots (2)$$

Now given that $PQ^2 = k^2$

$$\text{we have } (x - x_1)^2 + y^2 = k^2 \quad \text{or } x - x_1 = \pm \sqrt{k^2 - y^2}$$

$$\text{Hence using (2), we obtain } y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2} \dots (3)$$

(3) is the required differential equation for such curves

$$\text{Now solving (3) we get, } \int \frac{-y dy}{\sqrt{k^2 - y^2}} = \int -dx$$

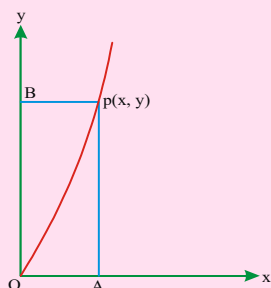
or $x^2 + y^2 = k^2$, which passes through $(0,k)$

W.E-12: A Curve $y=f(x)$ passes through the origin. Through any point (x,y) on the curve, lines are drawn parallel to the coordinate axes. If the curve divides the area formed by these lines and coordinates axes in the ratio $m:n$, find the curve.

Sol: Area of $OBPO$:area of $OPAP$ = $m:n$

$$\Rightarrow \frac{xy - \int_0^x y dx}{\int_0^x y dx} = \frac{m}{n} \Rightarrow nxy = (m+n) \int_0^x y dx$$

Differentiating with respect to x , we get



$$\begin{aligned} n(x \frac{dy}{dx} + y) &= (m+n)y \\ \Rightarrow nx \frac{dy}{dx} &= my \Rightarrow \frac{m}{n} \frac{dx}{x} = \frac{dy}{y} \\ \Rightarrow y &= cx^{m/n} \end{aligned}$$

➤ **Standard forms to solve differential equations:**

$$\text{i) } d\left(\frac{x^2}{y}\right) = \frac{2xydx - x^2dy}{y^2}$$

$$\text{ii) } d\left(\frac{y^2}{x}\right) = \frac{2xydy - y^2dx}{x^2}$$

$$\text{iii) } d\left(\frac{y^2}{x^2}\right) = \frac{2x^2ydy - 2xy^2dx}{x^4}$$

$$\text{iv) } d\left(\frac{x^2}{y^2}\right) = \frac{2y^2xdx - 2yx^2dy}{y^4}$$

$$\text{v) } d\left(\frac{1}{xy}\right) = \frac{-(xdy + ydx)}{x^2y^2}$$

$$\text{vi) } d(x^2 + y^2) = 2(xdx + ydy)$$

$$\text{vii) } d(\ln(x+y)) = \frac{dx + dy}{x+y}$$

$$\text{viii) } d(x^n y^n) = x^{n-1} y^{n-1} (nydx + nxdy)$$

W.E-13: The solution of the differential

$$\text{equation } x dx + y dy + \frac{xdy - ydx}{x^2 + y^2} = 0 \text{ is}$$

$$\text{Sol: We have, } x dx + y dy + \frac{xdy - ydx}{x^2 + y^2} = 0$$

$$\Rightarrow \frac{1}{2} d(x^2 + y^2) + d\left(\tan^{-1} \frac{y}{x}\right) = 0$$

On integrating, we obtain

$$\frac{1}{2} (x^2 + y^2) + \tan^{-1} \frac{y}{x} = \frac{c}{2}$$

$$\Rightarrow \frac{c - x^2 - y^2}{2} = \tan^{-1} \frac{y}{x} \Rightarrow y = x \tan\left(\frac{c - x^2 - y^2}{2}\right)$$

GENERAL SOLUTION	DIFFERENTIAL EQUATION
$y = A_1 e^{\alpha_1 x} + A_2 e^{\alpha_2 x}$ (A_1, A_2 are arbitrary constants)	$y_2 - (\alpha_1 + \alpha_2) y_1 + \alpha_1 \alpha_2 y = 0$
$y = A_1 e^{\alpha_1 x} + A_2 e^{\alpha_2 x} + A_3 e^{\alpha_3 x}$ (A_1, A_2, A_3 are arbitrary constants)	$y_3 - (\alpha_1 + \alpha_2 + \alpha_3) y_2 + (\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_3 \alpha_1) y_1 - \alpha_1 \alpha_2 \alpha_3 y = 0$
$y = e^{\alpha x} (A_1 \cos \beta x + A_2 \sin \beta x)$ (A_1, A_2 are arbitrary constants)	$y_2 - 2\alpha y_1 + (\alpha^2 + \beta^2) y = 0$
$y = A_1 x^m + A_2 x^n$ (A_1, A_2 are arbitrary constants)	$x^2 y_2 - (m + n - 1) x y_1 + m n y = 0$
$y = e^{\alpha x} [A x^2 + B x + C]$ $y = a \cos (m x + b)$ (a, b are arbitrary constants)	$y_3 - 3\alpha y_2 + 3\alpha^2 y_1 - \alpha^3 y = 0$ $y_2 + m^2 y = 0$

Where A_1, A_2, A_3 are arbitrary constants $y_1 = \frac{dy}{dx}, y_2 = \frac{d^2 y}{dx^2}, \dots$

DIFFERENTIAL EQUATION	SOLUTION
$x dy + y dx = 0$	$xy = C$
$\frac{y dx - x dy}{y^2} = 0$	$\frac{x}{y} = C$
$\frac{x dy - y dx}{x^2} = 0$	$\frac{y}{x} = C$
$\frac{y e^x dx - e^x dy}{y^2} = 0$	$\frac{e^x}{y} = C$
$\frac{x e^y dy - e^y dx}{x^2} = 0$	$\frac{e^y}{x} = C$
$\frac{x dy - y dx}{x^2 + y^2} = 0$	$\tan^{-1} \frac{y}{x} = C$

DIFFERENTIAL EQUATION	SOLUTION
$\frac{y dx - x dy}{x^2 + y^2} = 0$	$\tan^{-1} \frac{x}{y} = C$
$\frac{x dy + y dx}{xy} = 0$	$\log xy = C$
$\frac{x dy - y dx}{xy} = 0$	$\log \frac{y}{x} = C$
$\frac{y dx - x dy}{xy} = 0$	$\log \frac{x}{y} = C$
$\frac{x dx + y dy}{x^2 + y^2} = 0$	$\log \sqrt{x^2 + y^2} = C$

C.U.Q

1. The order of the differential equation of which $y^2 = 4ax$ is a solution, is
1. 2 2. 1 3. 3 4. Can't be determined
2. The differential equation of all parabolas whose axis of symmetry is parallel to y-axis is of order
1. 1 2. 2 3. 3 4. 4
3. The order of the differential equation whose general solution is given by $(x-a)^2 + (y-b)^2 = a^2$ is
1. 4 2. 3 3. 2 4. 1
4. The order and degree of the differential equation of rectangular hyperbola whose asymptotes are axes is
1. 2,2 2. 3,2 3. 1,2 4. 1,1
5. The order of D.E, whose solution is $y = C_1 e^{x+c_2} + C_3 \sin(x+c_4)$
1. 4 2. 2 3. 3 4. 1
6. The order of differential equation whose solution is given by $y = c_1 \cos(2x+c_2) - (c_3+c_4)a^{x+c_5} + c_6 \sin(x-c_7)$ is
1. 4 2. 3 3. 7 4. 5
7. $\frac{dy}{dx} = \frac{4x+2y+1}{x-2y+3}$ is a differential equation of the type
1. variable separable 2. exact
3. Homogeneous 4. non homogeneous
8. The type of the differential equation $x \frac{dy}{dx} + y = y^2 \log x$ is
1. Linear 2. Bernoulli
3. Exact 4. Homogenous
9. The type of D.E $(1+y^2)dx = (\tan^{-1} y - x)dy$ is
1. Linear in x 2. Bernoulli
3. Linear in y 4. Homogenous
- 10) Solution of $x dy = \left(\frac{f(xy)}{x f'(xy)} - y \right) dx$ is
1) $f(xy) = c$ 2) $x f(xy) = c$
3) $y f(xy) = c$ 4) $cx = f(xy)$

- 11) The differential equation of all straight lines which are at a constant distance from the origin is

1) $(y - xy_1)^2 = p^2(1 + y_1^2)$

2) $(y + xy_1)^2 = p^2(1 - y_1^2)$

3) $(y + xy_1)^2 = p^2(1 + y_1^2)$

4) $(y - xy_1)^2 = p^2(1 - y_1^2)$

- 12) If $f(x), g(x)$ and $h(x)$ are continuous, thrice differentiable functions and each satisfy the

differential equation $\frac{d^2 y}{dx^2} + y = 0$ then

$$\phi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f_1(x) & g_1(x) & h_1(x) \\ f_2(x) & g_2(x) & h_2(x) \end{vmatrix}$$

(here $f_k(x), g_k(x)$ and $h_k(x)$ are k^{th} derivatives of $f(x), g(x)$ and $h(x)$ respectively)

1) 0 2) $f(x)g(x)h(x)$

3) $f_1(x)g_1(x)h_1(x)$ 4) 1

- 13) If $\phi(x) = \phi^1(x)$ and $\phi(1) = 2$ then $\phi(3) =$

1) e^2 2) e^3 3) $3e^2$ 4) $2e^2$

- 14) The differential equation obtained by eliminating A, B from $y = A \cos wt + B \sin wt$ is _____

1) $y^{11} + y^1 = 0$ 2) $y^{11} - w^2 y = 0$

3) $y^{11} + w^2 y = 0$ 4) $y^{11} + y = 0$

- 15) The order and degree of the differential

equation $\left(\frac{d^3 y}{dx^3} \right)^{1/3} + \left(\frac{d^2 y}{dx^2} \right)^{2/3} + \frac{dy}{dx} = 0$ are

1) 3, 3 2) 3, 6 3) 3, 9 4) 4, 4

- 16) The general solution of

$$\left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right) y dx + \left(x \cos \frac{y}{x} - y \sin \frac{y}{x} \right) x dy = 0 \text{ is}$$

1) $xy \sin \frac{y}{x} = c$

2) $xy \cos \frac{y}{x} = c$

3) $\cos \frac{y}{x} = cxy$

4) $\sin \frac{y}{x} = cxy$

C.U.Q - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 2 | 2) 3 | 3) 3 | 4) 4 | 5) 3 | 6) 4 |
| 7) 4 | 8) 2 | 9) 1 | 10) 4 | 11) 1 | 12) 1 |
| 13) 4 | 14) 3 | 15) 1 | 16) 2 | | |

C.U.Q - HINTS

- Order = No of arbitrary constants.
- Equation of the parabola is
 $(x - h)^2 = \pm 4a(y - k)$
- Order = No of arbitrary constants.
- Rectangular hyperbola is $xy = c^2$
- Order = No of arbitrary constants.
- No. of arbitrary constants = 5
- General equation of the non homogeneous differential equation is

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$$
- General equation of the Bernoulli differential equation is $\frac{dy}{dx} + Py = Qy^n$, where P, Q functions in x
- Linear differential equation of the form in x is
 $\frac{dx}{dy} + px = Q$, where p, q are functions in y
- $x dy + y dx = \frac{f(xy)}{xf'(xy)} dx$; $d(xy) = \frac{f(xy)}{xf'(xy)} dx$

$$\Rightarrow \int \frac{f'(xy)}{f(xy)} d(xy) = \int \frac{dx}{x}$$
- Let the straight line be $x \cos \alpha + y \sin \alpha = P$ where α is arbitrary constant differentiate w.r.t 'x'
 ----- 1
 $\cos \alpha + \sin \alpha y_1 = 0$; $\tan \alpha = -\frac{1}{y_1}$ ----- 2
 from 1 and 2 $x \left(\frac{-y_1}{\sqrt{1+y_1^2}} \right) + y \left(\frac{1}{\sqrt{1+y_1^2}} \right) = P$
 $(y - xy_1)^2 = P^2(1 + y_1^2)$
- Differentiate and use $\frac{d^2y}{dx^2} + y = 0$

$$13. \frac{\phi'(x)}{\phi(x)} = 1$$

$$\int \frac{\phi'(x)}{\phi(x)} dx = \int 1 \cdot dx$$

$$\log[\phi(x)] = x + c$$

$$\text{Put } x = 1$$

$$\log[\phi(1)] = 1 + c$$

$$\log(2) = 1 + c$$

$$\therefore \log[\phi(x)] = x + \log(2) - 1$$

$$\text{put } x = 3$$

$$\phi(3) = e^{3+\log 2-1}$$

$$\phi(3) = e^{2+\log 2} = 2e^2$$

14. Diff two times and eliminate constants

15. Eliminate radicals, $a+b+c=0$

$$\Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$16. x \cos \frac{y}{x} (y dx + x dy) + y$$

$$\sin \frac{y}{x} (y dx - x dy) = 0$$

$$x \cos \frac{y}{x} d(xy) + y \sin \frac{y}{x} (y dx - x dy) = 0$$

$$x d(xy) = x^2 y \tan \left(\frac{y}{x} \right) d \left(\frac{y}{x} \right)$$

$$\frac{d(xy)}{xy} = \tan \left(\frac{y}{x} \right) d \left(\frac{y}{x} \right)$$

$$\Rightarrow \log(xy) = \log \sec \left(\frac{y}{x} \right) + \log c$$



LEVEL-I (C.W)

BASED ON ORDER AND DEGREE OF THE D.E

1. Order of $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 + 3y = x^4$ is
 1. 3 2. 1 3. 2 4. 4
2. If m and n are order and degree of the equation

$$\left(\frac{d^2y}{dx^2}\right)^5 + 4\left(\frac{d^2y}{dx^2}\right)^3 + \frac{d^3y}{dx^3} = x^2 - 1 \text{ then } (m, n) =$$

- 1) (3, 3) 2) (3, 2) 3) (3, 5) 4) (3, 1)
3. Order and degree of the differential equation

$$\frac{dy}{dx} + 2\frac{dx}{dy} = 7 \text{ is}$$

1. 1,1 2. 1,2 3. 2,1 4. 2,2
4. The degree of differential equation

$$x = 1 + \left(\frac{dy}{dx}\right) + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \dots \text{ is}$$

- 1) three 2) one
 3) not defined 4) four
5. The degree of differential equation of all circles in the first quadrant touching the coordinate axis is

1. 2 2. 1 3. 3 4. 4
6. The order of the differential equation

$$\text{satisfying } \sqrt{1-x^4} + \sqrt{1-y^4} = a(x^2 - y^2) \text{ is}$$

1. 1 2. 2 3. 3 4. 4
7. The differential equation representing the family curves $y^2 = 2c(x + \sqrt{c})$ when 'c' is a parameter is of

1. degree 4 2. order 2
 3. degree 3 4. degree 1
8. The differential equation of all conics with centre at origin is of order

- 1) 2 2) 3 3) 4 4) 1

9. The degree and order of the differential equation of all tangent lines to the parabola $x^2 = 4y$ is

- 1) 2, 1 2) 2, 2 3) 1, 3 4) 1, 4

10. The degree of differential equation of which $y^2 = 4a(x + a)$ is a solution is

1. 2 2. 3 3. 1 4. 4

BASED ON FORMATION OF DIFFERENTIAL EQUATIONS

11. If $y = e^{4x} + 2e^{-x}$ satisfies the differential equation $y_3 + Ay_1 + By = 0$ then

- 1) A = 12, B = 13 2) A = 13, B = 12
 3) A = -12, B = -13 4) A = -13, B = -12

12. D.E of $y = Ae^x + Be^{2x} + Ce^{3x}$ is

1. $y_3 - 6y_2 + 11y_1 - 6y = 0$
 2. $y_3 + 6y_2 + 11y_1 + 6y = 0$
 3. $y_3 - 6y_2 + 11y_1 + 6y = 0$
 4. $y_3 - 6y_2 - 11y_1 - 6y = 0$

13. D.E whose solution is

$$y = e^{3x}(c_1 \cos x + c_2 \sin x)$$

- 1) $y_2 - 3y_1 + 5y = 0$ 2) $y_2 - 6y_1 + 10y = 0$
 3) $y_2 + 6y_1 + 10y = 0$ 4) $y_2 - 6y_1 - 10y = 0$

14. The D.E whose solution is $y = c \sin x$ is

1. $\frac{dy}{dx} = y \cot x$ 2. $y_2 + y = 0$
 3. $\frac{dy}{dx} = y \cos x$ 4. $\frac{dy}{dx} = y \cos ecx$

15. The D.E whose solution is $y = ax^2 + bx + c$ is

1. $y_3 = y$ 2. $y_3 = x$ 3. $y_3 = 0$ 4. $y_2 = 0$

16. The D.E whose solution is $y^2 = 3ay - x^3$ is

1. $(x^3 - y^2)\frac{dy}{dx} = 3x^2y$ 2. $(x^3 - y^2)\frac{dy}{dx} = 3xy$
 3. $(x^3 - y)\frac{dy}{dx} = 3xy^2$ 4. $(y^2 - x^3)\frac{dy}{dx} = 3xy$

17. The D.E of $xy = ae^x + be^{-x} + x^2$ is

1. $xy_2 + y_1 + xy = 0$ 2. $xy_2 + 2y_1 = xy + 2 - x^2$
 3. $xy_2 + 2y = xy$ 4. $xy_2 + 2y_1 + xy + 2 = 0$

18. The differential equation of family of circles $x^2 + y^2 + 2\lambda x + c = 0$ is (where λ, c both are parameters)

1) $y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 1 = 0$ 2) $y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$
 3) $\left(\frac{d^2 y}{dx^2}\right) + \left(\frac{dy}{dx}\right)^2 = 0$ 4) $\frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 1 = 0$

BASED ON VARIABLE SEPARABLE

19. The Solution of $\frac{dy}{dx} = \frac{1-y}{1-x}$ is

1. $(1-x)(1+y) = c$ 2. $\frac{1-x}{1-y} = c$
 3. $(1-x)(1-y) = c$ 4. $\frac{1+x}{1-y} = c$

20. The solution of $\frac{dy}{dx} + y \tan x = 0$ is

1. $y = a \cos x$ 2. $y = a \sin x$
 3. $y = \log \cos x + c$ 4. $y = a \tan x + c$

21. The solution of $\frac{dy}{dx} = \frac{1+y^2}{\sec x}$ is

1. $\tan^{-1} y = \cos x + c$ 2. $\tan^{-1} y = \sin x + c$
 3. $y^3 = c \sec x + c$
 4. $\log(1+y^2) = \cos x + c$

22. The solution of $\frac{dy}{dx} = e^{\log x}$ is

1. $2y = x^2 + c$ 2. $y = x^2 + c$
 3. $y^2 = x + c$ 4. $xy = x^2 + c$

23. The solution of $\frac{dy}{dx} = \frac{x+x^2}{y-y^2}$ is

1. $3(y^2 - x^2) = 2(x^3 + y^3) + c$
 2. $3(x^2 + y^2) = 2(x^3 + y^3) + c$
 3. $x^2 - y^2 = x^3 + y^3 + c$
 4. $x^2 + y^2 = x^3 + y^3 + c$

24. The solution of $(1+e^x)ydy = e^x dx$ is

1. $y^2 = \log c(e^x + 1)$ 2. $\frac{y^2}{2} = \log ce^x$
 3. $\frac{y^2}{2} = \log c(e^x + 1)$ 4. $2y = \log c(e^x + 1)$

25. The solution of $ydx - xdy = 0$ is

1. $y^2 = cx$ 2. $y = cx^2$ 3. $y = cx$ 4. $x^2 = cy$

26. The solution of the D.E

$ydx - xdy + 3x^2 y^2 e^{x^3} dx = 0$ is

1. $x + e^{x^3} = cy$ 2. $x + ye^{x^3} - cy = 0$
 3. $x - ye^{x^3} + c = 0$ 4. $x + ye^{x^3} + cy^2 = 0$

27. When $x = 0, y = -1$ the particular solution of

the D.E $(1+y^2)dx + (1+x^2)dy = 0$ is

1. $x + y + xy = 1$ 2. $x + y + 1 = xy$
 3. $x + y + xy + 1 = 0$ 4. $x - y + xy = 1$

28. The solution of $e^x \tan y dx + (1-e^x) \sec^2 y dy = 0$ is

1. $\tan y = c(1-e^x)$ 2. $\sec y = c(1-e^x)$
 3. $\tan y(1-e^x) = c$ 4. $\sec y = 1-e^x$

29. The solution of

$3e^x \cos^2 y dx + (1-e^x) \cot y dy = 0$

1. $\tan y = c(e^x - 1)$ 2. $\tan^2 y = (e^x - 1)c$
 3. $\cot y = c(e^x - 1)^2$ 4. $\tan y = c(e^x - 1)^3$

30. Solution of the differential equation

$\frac{y + \sin x \cdot \cos^2(xy)}{\cos^2(xy)} dx + \left(\frac{x}{\cos^2(xy)} + \sin y \right) dy = 0$ is

1) $\tan(xy) - \cos x - \cos y = c$
 2) $\tan(xy) + \cos x - \cos y = c$
 3) $\tan(xy) + \cos x + \cos y = c$
 4) $\tan(xy) - \cos x + \cos y = c$

31. The population (x) in any year (t) is such that the rate of increase is proportional to the population then

1) $x = ct$ 2) $x = ce^{kt}$ 3) $x = ce^{-kt}$ 4) $x = c$

BASED ON HOMEGENOUS D.E

32. The solution of the differential equation

$$x^2 dy + y(x+y)dx = 0 \text{ is}$$

1) $x^2 y = c^2(2x+y)$ 2) $xy^2 = c^2(2x+y)$

3) $x^2 y = c^2(x+2y)$ 4) $xy^2 = c^2(x+2y)$

33. The solution of $\frac{dy}{dx} = \frac{y}{x} + \sin\left(\frac{y}{x}\right)$ is

1. $\tan\left(\frac{y}{2x}\right) - cx = 0$ 2. $\tan\left(\frac{y}{x}\right) - cx = 0$

3. $\tan\left(\frac{y}{x}\right) + cx - x = 0$ 4. $\sin\left(\frac{y}{2x}\right) - cx = 0$

34. Solution of $y^2 dx + (x^2 - xy + y^2) dy = 0$

1. $\tan^{-1}\left(\frac{x}{y}\right) + \log y + c = 0$

2. $2\tan^{-1}\left(\frac{x}{y}\right) + \log x + c = 0$

3. $\log\left(y + \sqrt{x^2 + y^2}\right) + \log y + c = 0$

4. $\log\left(y - \sqrt{x^2 + y^2}\right) - \log y + c = 0$

BASED ON NON-HOMEGENOUS D.E

35. Solution of $\frac{dy}{dx} = \frac{x-2y+3}{2x+y-5}$

1) $x^2 - y^2 - 4xy + 6x + 10y = c$

2) $x^2 + y^2 + xy + x - 3y = c$

3) $x^2 - y^2 + 2xy - x + 3y = c$

4) $x^2 + y^2 - xy + x - 3y = c$

36. The substitution required to change

$$(3y-7x+7)dx + (7y-3x+3)dy = 0 \text{ to a}$$

homogeneous differential equation is

1. $x = X+1, y = Y$ 2. $x = X, y = Y+1$

3. $x = X+1, y = Y+1$

4. $x = X+2, y = Y+2$

INTEGRATING FACTORS OF DIFFERENTIAL EQUATION

37. The Integrating Factor of the differential

$$\text{equation } (1-y^2)\frac{dx}{dy} + yx = ay \quad (-1 < y < 1)$$

1. $\frac{1}{y^2-1}$ 2. $\frac{1}{\sqrt{y^2-1}}$ 3. $\frac{1}{1-y^2}$ 4. $\frac{1}{\sqrt{1-y^2}}$

38. I.F of $x \log x \frac{dy}{dx} + y = 2 \log x$

1. x 2. $\log x$ 3. $\log(\log x)$ 4. $\frac{2}{x}$

39. I.F of $(1+x^2)\frac{dy}{dx} + y = e^{\tan^{-1}x}$

1) $\tan^{-1}x$ 2) $e^{\tan^{-1}x}$ 3) $\frac{1}{1+x^2}$ 4) $\tan x$

40. I.F of $\frac{dy}{dx} = \frac{x+y+1}{x+1}$

1) $\frac{1}{y+1}$ 2) $\frac{1}{x+1}$ 3) $\log(x+1)$ 4) $\log(y+1)$

41. I.F of $\frac{dy}{dx} - y \cot x = \operatorname{cosec} x$ is

1. $\sin x$ 2. $\tan x$ 3. $\sec x$ 4. $\operatorname{cosec} x$

LINEAR DIFFERENTIAL EQUATIONS

42. Solution of $\frac{dy}{dx} + 2y = e^{-x}$ is

1. $ye^x = 1 + ce^{-x}$ 2. $ye^{-x} = 1 + ce^{-x}$

3. $yx = e^{-x} + c$ 4. $e^x y = e^{-x} + ce^{-2x}$

BERNOULLIS EQUATION

43. The transformed equation of

$$\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$$

1) $\frac{du}{dx} + \frac{u}{x^2} = x$

2) $\frac{du}{dx} + ux = \frac{x^3}{2}$

3) $\frac{du}{dx} + 2ux = x^3$

4) $\frac{du}{dx} + \frac{u}{x} = x^2$

LEVEL-I (C.W.) - KEY

1) 3	2) 2	3) 2	4) 2	5) 1	6) 1
7) 3	8) 2	9) 1	10) 1	11) 4	12) 1
13) 2	14) 1	15) 3	16) 1	17) 2	18) 1
19) 2	20) 1	21) 2	22) 1	23) 1	24) 3
25) 3	26) 2	27) 2	28) 1	29) 4	30) 1
31) 2	32) 1	33) 1	34) 1	35) 1	36) 1
37) 4	38) 2	39) 2	40) 2	41) 4	42) 1
					43) 3

LEVEL-I (C.W.) - HINTS

- Order = highest derivative involved in the equation
- Standard
- $\frac{dy}{dx} + 2\frac{1}{dy} = 7$
- $x = e^{\left(\frac{dy}{dx}\right)} \Rightarrow \frac{dy}{dx} = \log x \therefore$ Order = 1, Degree = 1
- Let 'r' be radius of the circle. Equation is $(x-r)^2 + (y-r)^2 = r^2$
- $\frac{dy}{dx} = \frac{x\sqrt{1-y^4}}{y\sqrt{1-x^4}}$
- $y\frac{dy}{dx} = c \Rightarrow \left(y - 2x\frac{dy}{dx}\right)^2 = 4y\left(\frac{dy}{dx}\right)^3$
- Consider the equation of the conic has $ax^2 + 2hxy + by^2 = 1$
- Consider the tangent line as $x = my + \frac{1}{m}$
- $y\frac{dy}{dx} = 2a \Rightarrow y = 2x\frac{dy}{dx} + y\left(\frac{dy}{dx}\right)^2$
- Differentiating thrice and eliminating y_2 we have $A = -13$, $B = -12$
- The D.E of $y = A_1e^{\alpha_1x} + A_2e^{\alpha_2x} + A_3e^{\alpha_3x}$ is $y_3 - (\alpha_1 + \alpha_2 + \alpha_3)y_2 + (\alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_3\alpha_1)y_1 - \alpha_1\alpha_2\alpha_3y = 0$
- Differentiate twice verify with options
- Find y_1 15. Find y_3
- Find y_1
- differentiate two times with respect to x
- $x^2 + y^2 + 2\lambda x + c \Rightarrow 2x + 2y\frac{dy}{dx} + 2\lambda = 0$

$$\therefore x^2 + y^2 - \left(2x + 2y\frac{dy}{dx}\right)x + c = 0$$

$$\Rightarrow y^2 - x^2 - 2xy\frac{dy}{dx} + c = 0$$

Again differentiating both sides w.r.t. to x , we get

$$2y\frac{dy}{dx} - 2x - 2xy\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 - 2y\frac{dy}{dx} = 0$$

$$\Rightarrow 1 + y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$$

$$19. \frac{dy}{1-y} = \frac{dx}{1-x}$$

$$20. \frac{dy}{y} + \tan x dx = 0$$

$$21. \frac{dy}{1+y^2} = \cos x dx$$

$$22. dy = x dx$$

$$23. (y - y^2)dy = (x + x^2)dx$$

$$24.. ydy = \frac{e^x}{1+e^x} dx$$

$$25. \frac{dx}{x} - \frac{dy}{y} = 0$$

$$26. \text{dividing with } y^2$$

$$27. \frac{dx}{1+x^2} + \frac{dy}{1+y^2} = 0$$

$$28. \int \frac{e^x}{1-e^x} dx + \int \frac{\sec^2 y}{\tan y} dy = c$$

$$29. \int \frac{3e^x}{1-e^x} dx + \int \frac{\sec^2 y}{\tan y} dy = c$$

30. The given differential equation can be written as

$$\frac{y dx + x dy}{\cos^2(xy)} + \sin x dx + \sin y dy = 0$$

$$\int \frac{y dx + x dy}{\cos^2(xy)} + \int \sin x dx + \int \sin y dy = 0$$

$$\tan(xy) - \cos x - \cos y = c$$

$$31. \frac{dx}{dt} \alpha x \Rightarrow \frac{dx}{dt} = kx \Rightarrow \frac{dx}{x} = k dt \Rightarrow \log \frac{x}{c} = kt$$

$$32. \text{Put } y = vx$$

33. Put $\frac{y}{x} = v$ 34. Put $y=vx$ 35. $b_1 = -a_2$

36. by verification

37. G.E $\frac{dx}{dy} + x \frac{y}{1-y^2} = \frac{ay}{1-y^2}$

$$\begin{aligned} \text{I.F} &= e^{\int p dy} = e^{-\frac{1}{2} \int \frac{-2y}{1-y^2} dy} = e^{-\frac{1}{2} \log|1-y^2|} \\ &= e^{\log \frac{1}{\sqrt{1-y^2}}} = \frac{1}{\sqrt{1-y^2}} \end{aligned}$$

38. dividing with $x \log x$

39. $I.F = e^{\int \frac{1}{1+x^2} dx}$

40. $\frac{dy}{dx} - \frac{y}{x+1} = 1$, $I.F = e^{\int \frac{-dx}{x+1}}$

41. $I.F = e^{\int -\cot x dx}$

42. $I.F = e^{\int 2 dx}$

solution is $y(I.F) = \int (I.F) e^{-x} dx + C$

43. Dividing with $\cos^2 y$ and put $\tan y = u$



1. Order of $(3x+2y)\left(\frac{dy}{dx}\right)^2 + 5x \frac{d^2y}{dx^2} = 0$ is

1. 2 2. 1 3. 4 4. 3

2. Order and degree of

$$\sqrt{1 + \left(\frac{d^2y}{dx^2}\right)^3} = \left(2 + \frac{dy}{dx}\right)^{1/3} \text{ is}$$

1. 2,3 2. 2,9 3. 2,6 4. 2,2

3. The order and degree of the differential

equation $\frac{d^2y}{dx^2} = \sin\left(\frac{dy}{dx}\right) + xy$ are respectively.

1) 2,1 3) 2, Infinite
2) 2,0 4) 2, not defined

4. Order of the differential equation of the family of all concentric circles centred at (h, k) is

1. 1 2. 2 3. 3 4. 4

5. The number of arbitrary constants in the solution of a differential equation of degree 3 and order 2 is

1. 2 2. 3 3. 5 4. 1

6. The degree of the differential equation of all curves having normal of constant length 'c' is

1) 1 2) 3 3) 4 4) 2

7. The degree and order of the differential equation of the family of all parabolas whose axis is X-axis are respectively

1) 2, 1 2) 1, 2 3) 3, 2 4) 2, 2

8. The differential equation of all conics with the co-ordinate axes as axes is of the order

1) 1 2) 2 3) 3 4) 4

9. Family $y = Ax + A^4$ of curves is represented by the differential equation of degree

1. 3 2. 2 3. 4 4. 1

BASED ON FORMATION OF DIFFERENTIAL EQUATIONS

10. The D.E whose solution is $y = a \cos(3x+b)$ is

1. $y_2 + 3y = 0$ 2. $y_2 + y = 0$
3. $y_2 + 9y = 0$ 4. $y_2 + 6y = 0$

11. The D.E whose solution is $y = (c_1x + c_2)e^{5x}$

1) $y_2 + 10y_1 + 24y = 0$ 2) $y_2 - 10y_1 + 25y = 0$
3) $y_2 - 5y_1 + 25y = 0$ 4) $y_2 - 5y_1 + 10y = 0$

12. The D.E whose solution in $y = ce^x$ is

1. $y_1 = y^2$ 2. $y_2 = y_1$ 3. $y_1 = y$ 4. $y_1 + y = 0$

13. The D.E whose solution is $y = A \cos x + \sin x$ is

1. $\frac{dy}{dx} + y \tan x = \sec x$

2. $\frac{dy}{dx} + y \sin x = \cos x$

3. $\frac{dy}{dx} + y \sec x = \tan x$

4. $\frac{dy}{dx} + y \cot x = \operatorname{cosec} x$

14. The D.E whose solution is

$$y = A \sin 2x + B \cos 2x \text{ is}$$

1. $y_2 = 4y$ 2. $y_2 + 3y = 0$

3. $y_2 + y = 0$ 4. $y_2 + 4y = 0$

15. Form the differential equation by eliminating A,B,C from

$$y = e^{3x} (A + Bx + Cx^2) \text{ is}$$

1. $\frac{d^3 y}{dx^3} + 9 \frac{d^2 y}{dx^2} - 27 \frac{dy}{dx} + 27y = 0$

2. $\frac{d^3 y}{dx^3} + 9 \frac{d^2 y}{dx^2} + 27 \frac{dy}{dx} + 27y = 0$

3. $\frac{d^3 y}{dx^3} - 9 \frac{d^2 y}{dx^2} - 27 \frac{dy}{dx} - 27y = 0$

4. $\frac{d^3 y}{dx^3} - 9 \frac{d^2 y}{dx^2} + 27 \frac{dy}{dx} - 27y = 0$

16. The differential equation of all non- horizontal lines in a plane is

1) $\frac{d^2 y}{dx^2} = 0$ 2) $\frac{d^2 x}{dy^2} = 0$

3) $\frac{dy}{dx} = 0$ 4) $\frac{dx}{dy} = 0$

17. The differential equation of the family of curves $r^2 = a^2 \cos 2\theta$ where 'a' is arbitrary constant is

1) $\frac{dr}{d\theta} = -r \tan 2\theta$ 2) $\frac{dr}{d\theta} = r \cot 2\theta$

3) $\frac{dr}{d\theta} \cos 2\theta - r \sin 2\theta = 0$ 4) $\frac{dr}{d\theta} = 0$

18. The differential equation of the family of circles with fixed radius 5 units and centre on the line $y=2$ is

1. $(x-2)(y')^2 = 25 - (y-2)^2$

2. $(y-2)(y')^2 = 25 - (y-2)^2$

3. $(y-2)^2 (y')^2 = 25 - (y-2)^2$

4. $(x-2)^2 (y')^2 = 25 - (y-2)^2$

BASED ON VARIABLE SEPARABLE

19. The solution of $\frac{dy}{dx} = \left(\frac{y}{x}\right)^{1/3}$ is

1. $x^{2/3} + y^{2/3} = c$ 2. $y^{2/3} - x^{2/3} = c$

3. $x^{1/3} + y^{1/3} = c$ 4. $y^{1/3} - x^{1/3} = c$

20. The solution of $\frac{dy}{dx} + \frac{\sqrt{1+y^2}}{\sqrt{1+x^2}} = 0$ is

1. $\sin^{-1} x + \sin^{-1} y = c$ 2. $\tan^{-1} x + \tan^{-1} y = c$

3. $\sinh^{-1} x + \sinh^{-1} y = c$

4. $\cot^{-1} x + \cot^{-1} y = c$

21. The solution of $(e^x + e^{-x}) \frac{dy}{dx} = e^x - e^{-x}$ is

1. $y = \log |e^x - e^{-x}| + c$ 2. $y = \sinh x + c$

3. $y = \log |e^x + e^{-x}| + c$ 4. $y = \operatorname{sech} x + c$

22. The solution of $\frac{dy}{dx} - \frac{2xy}{1+x^2} = 0$ is

1. $y = c(1+x^2)$ 2. $y = c\sqrt{1+x^2}$

3. $y = \frac{c}{1+x^2}$ 4. $y = \frac{c}{\sqrt{1+x^2}}$

23. The solution of $\frac{dy}{dx} = e^{x-y}$ is

1. $e^x - e^y = c$

2. $e^x + e^y = c$

3. $e^{-x} - e^y = c$

4. $e^{-x} - e^{-y} = c$

24. The solution of $(1+x^2) \frac{dy}{dx} = 2x \cot y$ is

1. $\sec y = c(1+x^2)$ 2. $\cos y = c(1+x^2)$

3. $\tan y = c(1+x^2)$ 4. $\cot y = c(1+x^2)$

25. The solution of $\frac{dy}{dx} = (1+y^2)(1+x^2)^{-1}$ is

1. $y - x = c(1+xy)$ 2. $y + x = c(1+xy)$

3. $y = (1+2x)c$ 4. $xy = x^2 + x + c$

26. Equation of the curve passing through

(1,3) having slope $\frac{y}{x}$ at any point is

1. $y = 3x$ 2. $y = 3x^2$ 3. $x^2 = 3y$ 4. $x = 3y$

27. The solution of $\frac{dy}{dx} = e^{x-y} + e^{2\log x - y}$

1. $e^y = e^x + \frac{x^2}{3} + c$ 2. $e^y = e^x + \frac{x^3}{3} + c$

3. $e^y = e^x + \log x + c$ 4. $y = \frac{e^{3x+y}}{3}$

28. The solution of $\tan x dy + \tan y dx = 0$

1. $\tan x \tan y = c$ 2. $\sec^2 x + \sec^2 y = c$

3. $\sin x \sin y = c$ 4. $\cot x \cdot \cot y = c$

29. $\sqrt{1+x^2} dx + \sqrt{1+y^2} dy = 0$

1. $x\sqrt{1+x^2} + y\sqrt{1+y^2} + \log \left[\left(x + \sqrt{1+x^2} \right) \left(y + \sqrt{1+y^2} \right) \right] = c$

2. $\sqrt{1+x^2} + \sqrt{1+y^2} = c$

3. $\frac{1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+y^2}} = c$

4. $\log \left\{ \left(\sqrt{1+x^2} \right) + \left(\sqrt{1+y^2} \right) \right\} = x + c$

30. The solution of $x\sqrt{1+y^2} dx + y\sqrt{1+x^2} dy = 0$

1) $\sinh^{-1} x + \sinh^{-1} y = c$

2) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$

3) $(1+x^2)(1+y^2) = c$

4) $\sqrt{\frac{1+x^2}{1+y^2}} = c$

BASED ON HOMOGENEOUS D.E

31. By substituting $x = vy$ the transformed

equation of $(1 + e^{x/y}) dx + e^{x/y} \left(1 - \frac{x}{y} \right) dy = 0$ is

1. $y \frac{dv}{dy} - \left(\frac{1+e^v}{1-e^v} \right) = 0$ 2. $y \frac{dy}{dv} + \left(\frac{v+e^v}{1+e^v} \right) = 0$

3. $y \frac{dv}{dy} + x(v+e^v) = 0$ 4. $y \frac{dy}{dv} + x(v+e^v) = 0$

32. The solution of $\frac{dy}{dx} = \frac{\sqrt{x^2 - y^2} + y}{x}$

1) $\tan^{-1} \left(\frac{y}{x} \right) = \log(cx)$ 2) $\sin^{-1} \left(\frac{y}{x} \right) = \log(cx)$

3) $\cos^{-1} \left(\frac{y}{x} \right) = \log(cy)$ 4) $\sec^{-1} \frac{y}{x} = \log(cy)$

BASED ON NON-HOMOGENEOUS D.E

33. Solution of $\frac{dy}{dx} = \frac{x-y+2}{x+y-1}$

1) $x^2 + y^2 + xy - 4y - 2x = c$

2) $x^2 - y^2 - 2xy + 4x + 2y = c$

3) $x^2 - y^2 + xy + 2x - 4y = c$

4) $x^2 + y^2 - xy + 4x - 2y = c$

34. The solution of the differential equations

$\frac{dy}{dx} = \frac{x-2y+1}{2x-4y}$ is (E-08)

1) $(x-2y)^2 + 2x = c$ 2) $(x-2y)^2 + x = c$

3) $(x-2y) + 2x^2 = c$ 4) $(x-2y) + x^2 = c$

INTEGRATING FACTORS OF DIFFERENTIAL EQUATION

35. I.F of $(1+x^3) \frac{dy}{dx} + 3x^2 y = \sin^2 x$

1. $1+x^3$ 2. $\frac{1}{1+x^3}$ 3. $3x^2$ 4. $\frac{1}{x^3}$

36. I.F of $\frac{dy}{dx} + y = \frac{y+1}{x}$

1. $\frac{x}{e^x}$ 2. $\frac{e^x}{x}$ 3. xe^x 4. e^x

37. I.F of $\cos^2 x \frac{dy}{dx} + y = \tan x$

1. $e^{\sec x}$ 2. $e^{\tan x}$ 3. $e^{\sin x}$ 4. $e^{\cos x}$

38. I.F of $\frac{dy}{dx} + y \sec x = \tan x$ is

1. $\sec x$ 2. $\tan x$ 3. $\sec x + \tan x$ 4. $\cos x$

39. If $\sin x$ is an integrating factor of the differential equation $\frac{dy}{dx} + Py = Q$, then P can be

- 1) $\log \sin x$ 2) $\cot x$ 3) $\sin x$ 4) $\log \cos x$

LINEAR DIFFERENTIAL EQUATIONS

40. Solution of $\frac{dy}{dx} + \frac{2y}{x} = 5x^2$

1. $x^2 y = \frac{x^3}{3} + c$ 2. $x^2 y = \frac{x^4}{4} + c$
3. $x^2 y = x^5 + c$ 4. $x^2 y = x^3 + c$

LEVEL-I (H.W.) - KEY

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1) 1 | 2) 2 | 3) 4 | 4) 1 | 5) 1 | 6) 4 | 7) 2 |
| 8) 2 | 9) 3 | 10) 3 | 11) 2 | 12) 3 | 13) 1 | 14) 4 |
| 15) 4 | 16) 2 | 17) 1 | 18) 3 | 19) 2 | 20) 3 | 21) 3 |
| 22) 1 | 23) 1 | 24) 1 | 25) 1 | 26) 1 | 27) 2 | 28) 3 |
| 29) 1 | 30) 2 | 31) 2 | 32) 2 | 33) 2 | 34) 1 | 35) 1 |
| 36) 2 | 37) 2 | 38) 3 | 39) 2 | 40) 3 | | |

LEVEL-I (H.W.) - HINTS

1. Order = highest derivative involved in the equation

2. $\left(\sqrt{1 + \left(\frac{d^2 y}{dx^2} \right)^3} \right)^6 = \left[\left(2 + \frac{dy}{dx} \right)^{1/3} \right]^6$

3. equation is not a polynomial in differential hence degree is not defined

Degree = Power of the highest derivative

4. Equation represent family of circles is

$$(x-h)^2 + (y-k)^2 = r^2$$

(r is arbitrary constant).

5. Order = Number of arbitrary constants

6. Order = highest derivative involved in the equation
Degree = Power of the highest derivative

7. Equation of the parabola whose axis is

$$X\text{-axis is given by } y^2 = 4a(x+h).$$

Where a, h are arbitrary constants. Hence order = 2.
On differentiating w.r.t x ,

$$\text{we get } 2y \frac{dy}{dx} = 4a \Rightarrow y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$$

8. Consider the equation of conic has $ax^2 + by^2 = 1$

9. $\frac{dy}{dx} = A \Rightarrow y = x \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^4$

10. Eliminate constants a and b

11. Differentiate twice verify with options

12. Find y_1 13. Find y_1 14. Find y_2

15. $y_3 - 3\alpha y_2 + 3\alpha^2 y_1 - \alpha^3 y = 0$

16. $x = my + c \quad \therefore \frac{d^2 x}{dy^2} = 0$

17. $r^2 = a^2 \cos 2\theta \Rightarrow r \frac{dr}{d\theta} = -a^2 \sin 2\theta$

$$\Rightarrow r \frac{dr}{d\theta} = -\frac{r^2}{\cos 2\theta} \cdot \sin 2\theta$$

18. Eq of circle $(x-a)^2 + (y-2)^2 = 25$

Diff w.r.t 'x' $a = x + (y-2)y_1$

19. $y^{-1/3} dy = x^{-1/3} dx$ 20. $\frac{dy}{\sqrt{1+y^2}} + \frac{dx}{\sqrt{1+x^2}} = 0$

21. $\int dy = \int \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) dx$

22. $\frac{dy}{y} - \frac{2xdx}{1+x^2} = 0$

23. $e^y dy = e^x dx$

24. $\tan y dy = \frac{2x}{1+x^2} dx$ 25. $\frac{dx}{1+x^2} = \frac{dy}{1+y^2}$

26. $\frac{dy}{dx} = \frac{y}{x}$

27. $e^y dy = (e^x + x^2) dx$

28. $\int \cot y dy + \int \cot x dx = c$

29. $\int \sqrt{1+x^2} dx + \int \sqrt{1+y^2} dy = c$

30. $\frac{2x}{\sqrt{1+x^2}} dx = \frac{-2y}{\sqrt{1+y^2}} dy$

31. $\frac{dx}{dy} + \frac{e^{x/y} \left(1 - \frac{x}{y} \right)}{1 + e^{x/y}} = 0$ 32. put $y = vx$

33. $xdy + ydy - dy = xdx - ydx + 2dx$ 34. put $x-2y=u$

35. dividing with $(1+x^3)$ 36. $\frac{dy}{dx} + y \left(1 - \frac{1}{x} \right) = \frac{1}{x}$

37. dividing with $\cos^2 x$ 38. I.F. = $e^{\int \sec x dx}$

39. I.F. = $\sin x, e^{\int p dx} = \sin x$

40. I.F. = $e^{\int \frac{2}{x} dx}$ solution is $y(I.F.) = \int (I.F.) 5x^2 dx + C$